

Discussion of "Single-Image Super-Resolution of Sentinel-2 Low Resolution Bands with Residual Dense Convolutional Neural Networks"

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Abstract. This report provides an analysis of the research paper by Salgueiro et al. 2021 on single image super-resolution technique for enhancing Sentinel-2 satellite images. The authors proposed Sen2-RDSR, a deep convolutional neural network architecture that simultaneously improve the 20m low resolution (LR) and 60m very low resolution (VLR) bands to achieve a 10m spatial high resolution (HR). The model has two-branch architecture, where the first branch processes 20m bands and the second branch leverages the super-resolved 20m outputs to enhance 60m bands. The experimental results demonstrate superior performance compared to the existing methods in six quality metrics, including RMSE, SRE, SAM, PSNR, SSIM, and ERGAS. The research tackle a critical limitation in Earth observation applications where high spatial resolution is essential for precise land cover classification and vegetation monitoring. This analysis examines the technical contributions, methodological approaches, and practical implications of the proposed solution. The report also discusses the assumptions underlying the approach, identifies potential limitations, evaluates the significance of this work for advancing remote sensing applications and small scale implementation of the proposed approach.

1 Introduction

The European Space Agency's Sentinel-2 mission represents a cornerstone of contemporary Earth observation, providing free access to multispectral satellite imagery with unprecedented global coverage and temporal revisit capabilities [Salgueiro et al., 2021]. These twin satellites capture imagery across 13 spectral bands with varying spatial resolutions: four high-resolution bands at 10 meters per pixel, six low-resolution bands at 20 meters per pixel, and three very low-resolution bands at 60 meters per pixel. While this multi-resolution approach optimizes data transmission and storage constraints, it creates significant challenges for applications requiring uniform high spatial resolution across all spectral channels.

The work by Salgueiro, Marcello, and Vilaplana addresses this fundamental limitation through the development of Sen2-RDSR (Sentinel-2 Residual Dense Super-Resolution), a deep learning-based approach for enhancing the spatial resolution of Sentinel-2's lower resolution bands, which is important for land-use application Figure 1. The research

tackles a problem of considerable practical importance, as many Earth observation applications, including precision agriculture, urban planning, and environmental monitoring, require detailed spatial information that can only be achieved through high-resolution imagery across all spectral bands.

The authors' approach represents a significant improvement from previous methodologies in several key aspects. Traditional super-resolution techniques for Sentinel-2 imagery typically employ separate models for different resolution groups, requiring independent processing pipelines for 20m and 60m bands. In contrast, Sen2-RDSR implements a unified architecture that processes both resolution groups simultaneously while leveraging the interdependencies between them.

The technical foundation of Sen2-RDSR builds upon Residual in Residual Dense Blocks (RRDB), a neural network component originally developed for natural image super-resolution. [Zhou et al., 2020] The authors adapt this architecture specifically for the characteristics of satellite imagery, incorporating domain-specific considerations such as spectral consistency and spatial coherence across multiple

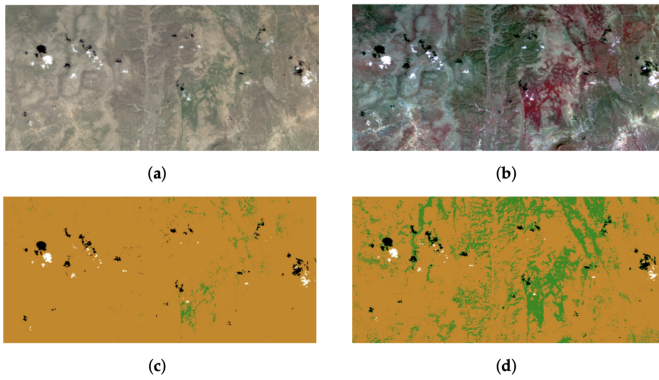


Figure 1. Comparison of segmentation maps obtained with only the 10 m GSD bands and with a combination of 10 m and SR 20 m bands. (a) True Color 10 m GSD (RGB); (b) False color 20 m GSD B7,B6,B5; (c) Segmentation using only 10 m bands; (d) Segmentation using 10 m and 20 m.[Salgueiro et al., 2021]

bands. The model architecture consists of two interconnected branches: the first branch enhances 20m resolution bands using both the original 20m data and the native 10m bands as input, while the second branch processes 60m bands using the original 60m data, the 10m bands, and the super-resolved 20m outputs from the first branch.

This architecture is mainly based on two key assumptions: first, all images or bands are self-similar, this means that the structural features in one band should be reflected on the other bands, the second assumption is that all images are in the same scale meaning that all images cover exactly the same earth surface area.

Figure 2 illustrates the complete Sen2-RDSR architecture, showing the two-branch design and the flow of information between different resolution inputs. The unified approach enables the model to leverage spatial information across multiple resolution scales while maintaining computational efficiency through shared feature extraction components.

This cascaded approach is the key innovation in the methodology. Rather than treating the super-resolution of different resolution groups as independent problems, the authors recognize that the enhanced 20m bands can provide valuable spatial information for improving the 60m band enhancement. This design choice reflects an understanding that spatial patterns and features often exhibit consistency across different spectral bands, allowing information from higher resolution outputs to guide the enhancement of lower resolution inputs.

The core building block of the Sen2-RDSR architecture is the Residual in Residual Dense Block (RRDB), which combines the advantages of residual learning and dense connections to enable effective feature extraction and reuse.[Kaiming H et al., 2015] [Gao H et al., 2018] As shown in Figure 3, each RRDB contains three dense blocks with local residual connections, allowing each layer to ac-

cess features from all preceding layers within the block. Each dense block is followed by scaling factor (X_b) for training stability.[Szegedy et al., 2017] The global residual connection around the entire RRDB facilitates gradient flow during training and helps preserve low-frequency information from the input.

The training methodology employed by the authors follows a two-stage approach; first, the model focuses exclusively on learning the mapping from 20m to 10m resolution using 8000 random patches of 32x32 pixels. Subsequently, the training incorporates the 60m to 10m enhancement objective using 500 random patches. In the context of supervised learning, ground-truth images are required but due data limitation problem the authors train the model using the original data as ground-truth and synthetically downsampled version as a input.

The experimental validation of Sen2-RDSR include both quantitative and qualitative assessments across multiple evaluation metrics. The authors employ six distinct quantitative measures: Root Mean Square Error (RMSE), Signal-to-Reconstruction Error Ratio (SRE), Spectral Angle Mapper (SAM), Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index Measure (SSIM), and Error Relative Global Dimensionless Error Sum (ERGAS). This comprehensive evaluation framework ensures that the assessment captures different aspects of image quality, from pixel-level accuracy to perceptual similarity, contrast, luminance and spectral fidelity.

The quantitative results demonstrate superior performance of Sen2-RDSR compared to existing state-of-the-art methods across most evaluation metrics. Figure 4 presents a comprehensive comparison showing the performance improvements achieved by Sen2-RDSR over baseline methods including bicubic interpolation, DSen2, and the approach by Zhang. [Lanaras et al. 2018] [Zhang et al., 2020] The results indicate consistent improvements in both 20m and 60m band enhancement tasks.

Beyond quantitative metrics, the visual quality assessment reveal improvements in spatial detail preservation and artifact reduction. Figure 6 shows representative examples of 20m band enhancement results, comparing the original low-resolution inputs, bicubic interpolation, competing methods, and Sen2-RDSR outputs. The enhanced imagery demonstrates improved edge sharpness, texture preservation, and overall visual quality.

The qualitative measurements are only for visual inspection to give a sense the proposed model is delivering a better super-resolved images than the previous state-of-the-art models.

2 Related Literature

The field of satellite image super-resolution has experienced rapid development in recent years, driven by the increasing

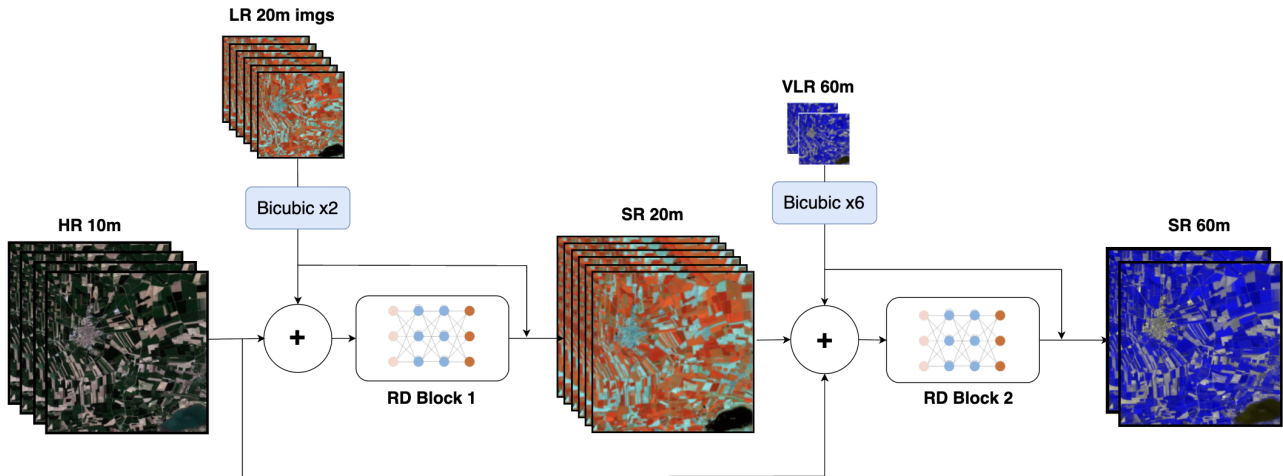


Figure 2. Sen2-RDSR model architecture showing the two-branch design for simultaneous enhancement of 20m and 60m bands. The first branch processes 10m and 20m inputs to generate super-resolved 20m outputs, while the second branch utilizes these outputs along with 10m and 60m inputs to produce enhanced 60m imagery.

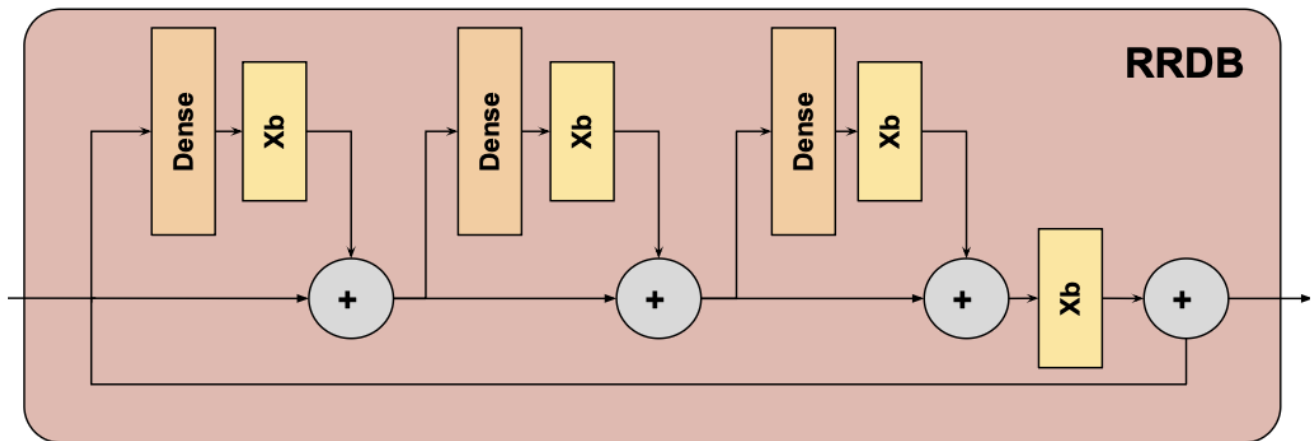


Figure 3. Internal structure of the Residual in Residual Dense Block (RRDB) showing dense connections within each dense block and the global residual connection. The residual scaling factor x_b controls the contribution of the dense block outputs to maintain training stability. [Salgueiro et al., 2021]

availability of satellite data and advances in deep learning techniques. This section examines key contributions in the domain, Two related to the work before the Sens-RDSR approach and the three posterior approaches.

2.1 Deep Learning Approaches for Remote Sensing Image Super-Resolution

The use of deep learning to remote sensing image super-resolution represents a new paradigm shift from traditional interpolation and reconstruction-based methods. Wang et al. (2022) provide a comprehensive review of deep learning-based single image super-resolution methods for optical re-

mote sensing images [Wang et al., 2022]. Their analysis reveals that while early approaches relied on simple convolutional neural networks, recent developments have embraced more sophisticated architectures including residual networks, attention mechanisms, and generative adversarial networks [Zhang et al. 2020].

Remote sensing images present distinct characteristics that differentiate them from natural images: they are captured from extreme distances, contain diverse land cover types with varying scales, and often exhibit complex spatial relationships between different spectral bands. These characteristics necessitate specialized approaches that can effectively handle the multi-scale nature of satellite imagery.

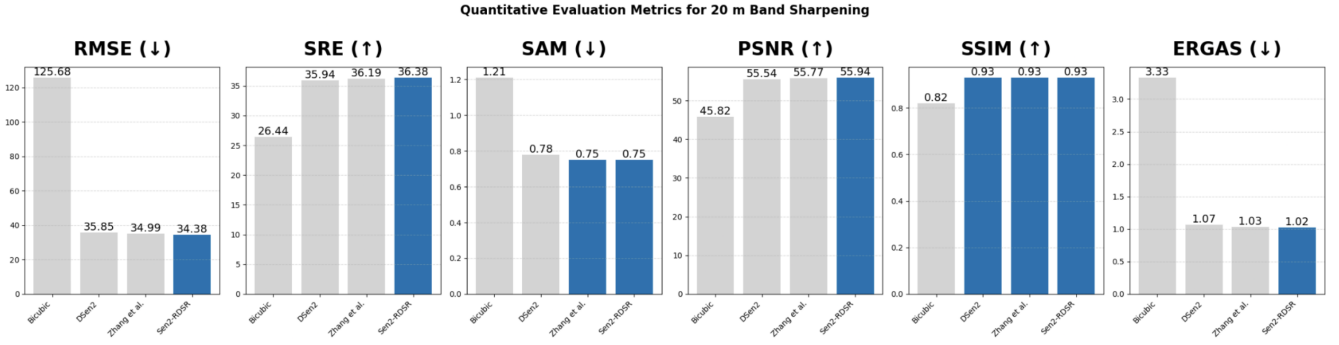


Figure 4. Quantitative comparison of Sen2-RDSR with state-of-the-art methods across six evaluation metrics (RMSE, SRE [dB], SAM, PSNR [dB], SSIM [dB], ERGAS) for 20m band. Lower values are better for RMSE, SAM, and ERGAS, while higher values are better for SRE, PSNR, and SSIM.

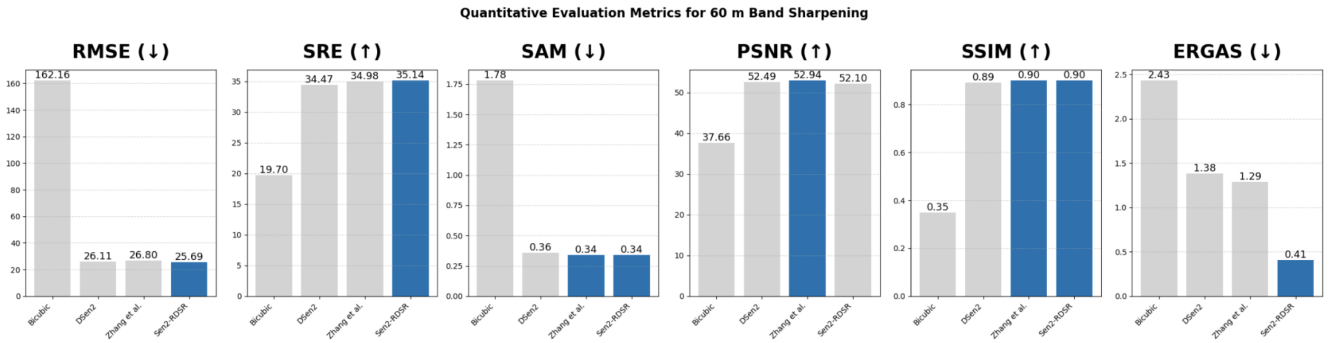


Figure 5. Quantitative comparison of Sen2-RDSR with state-of-the-art methods across six evaluation metrics (RMSE, SRE [dB], SAM, PSNR [dB], SSIM, ERGAS) for 60m band. Lower values are better for RMSE, SAM, and ERGAS, while higher values are better for SRE, PSNR, and SSIM.

2.2 Learning a globally applicable deep neural network

The introduction of skip connections or residual connections is widely used for computer vision tasks and in this approach on image super-resolution have shown to out perform analytical methods. [Lanaras et al. 2018] proposed a CNN with skip connections between the features maps also called resblocks [Kaiming H et al., 2015]. Using two separated networks to super-resolve both 20m and 60 bands. The LR bands are first upsampled with bicubic interpolation and concatenated with HR bands before entering the network.

2.3 High performance distributed deep learning

The use of resblocks [Kaiming H et al., 2015] with self-attention mechanism [Zhang et al., 2020] together with a proposed distributed training procedure suitable for super computers, make this approach to reach a state-of-the-art results, speeding-up the training process while maintaining the loss of performance at minimum values for both models. This contribution use two different models to enhance the VLR and LR images. For comparison purpose the Sen2-RDSR

model use exactly the same training data as the proposed by Zhang et al.

2.4 Attention Mechanisms and Spectral Considerations

Zabalza and Bernardini (2022) introduce a significant advancement in satellite image super-resolution through their work on spectral attention mechanisms [Zabalza and Bernardini, 2022]. Their research addresses the specific challenge of enhancing Sentinel-2 imagery from 10m to 2.5m resolution using a Spectral Attention Residual Network (SARNet). The incorporation of Residual Channel Attention Blocks (RCAB) represents a crucial innovation that allows the network to selectively focus on the most informative spectral channels during the enhancement process.

The spectral attention mechanism proposed by Zabalza and Bernardini addresses the limitation of traditional super-resolution approaches that treat all spectral channels equally. In satellite imagery, different spectral bands contain varying amounts of spatial information, and their relative importance can change depending on the specific land cover types present in the scene. The attention mechanism enables

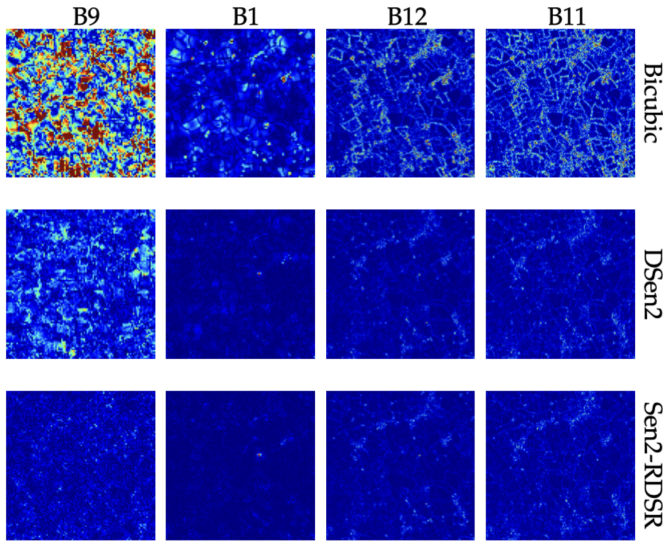


Figure 6. Qualitative comparison of 20m band enhancement results. From top to bottom: original 20m input minus bicubic interpolation, DSen2 result and Sen2-RDSR result. The Sen2-RDSR output shows superior detail preservation and reduced blurring artifacts, while darker blue are better. [Salgueiro et al., 2021]

the network to adaptively weight different spectral channels based on their contribution to the spatial enhancement task.

2.5 Real-World Degradation Modeling

Zhang et al. (2022) address a gap in satellite image super-resolution research through their focus on real world degradation modeling [Zhang et al., 2022]. Their work recognizes that most existing approaches rely on synthetic training data created through simple downsampling operations, which fail to capture the complex degradation processes that occur during actual satellite image acquisition. The authors propose a more realistic training framework that models blur kernels and imaging noise to better represent the conditions encountered in operational satellite systems.

The degradation modeling approach introduced by Zhang et al. represents a methodological advancement that bridges the gap between laboratory performance and real-world applicability. Traditional super-resolution methods often achieve excellent results on synthetic test data but experience substantial performance degradation when applied to actual satellite imagery.

3 Discussion

The work by Salgueiro et al. represents a interesting contribution to the field of satellite image super-resolution, demonstrating both technical innovation and practical utility. However, a critical examination of their approach reveals sev-

eral important considerations regarding the methodology, assumptions, and broader implications of their research.

3.1 Technical Contributions and Novelty

The primary technical contribution of Sen2-RDSR lies in its unified architecture that simultaneously processes both 20m and 60m resolution bands rather than treating them as independent enhancement tasks. This design choice reflects an understanding of the interdependencies between different resolution groups in Sentinel-2 imagery. The cascaded approach, where super-resolved 20m bands inform the enhancement of 60m bands, represents a logical exploitation of the hierarchical nature of spatial information in multispectral satellite data.

The adaptation of Residual in Residual Dense Blocks (RRDB) to satellite imagery demonstrates the authors' ability to transfer successful architectures from natural image processing to the remote sensing domain. However, the paper provides limited discussion of how the specific characteristics of satellite imagery influenced their architectural choices. The spectral consistency requirements, radiometric accuracy preservation, and computational constraints associated with large satellite images all present unique challenges that may require domain-specific modifications to standard computer vision architectures.

3.2 Methodological Assumptions and Limitations

Several implicit assumptions underlying the Sen2-RDSR approach deserve critical examination. First, the authors assume that spatial patterns and features exhibit sufficient consistency across different spectral bands to justify the cascaded enhancement strategy. While this assumption appears reasonable for many land cover types, it may not hold universally across all spectral regions and geographical contexts.

The training data used in the study appears to be geographically constrained, though the authors do not provide detailed information about the spatial distribution or diversity of their training samples. This limitation raises important questions about the generalization of the trained model to different climatic zones, land cover types, and seasonal conditions.

3.3 Model Complexity and computational costs

One important aspect of using deep learning models for image processing tasks is their complexity, in terms of the number of model parameters (weights) and the associated computational cost. Although this research does not mention it explicitly, these factors are critical for deploying models in production. Computational cost translates directly into financial cost, which is decisive for large-scale applications.

3.4 Spectral Fidelity and Radiometric Accuracy

One of the most critical aspects of satellite image enhancement concerns the preservation of spectral fidelity and radiometric accuracy. Many remote sensing applications, including atmospheric correction, surface reflectance analysis, and quantitative vegetation monitoring, depend on precise radiometric measurements. The paper employs Spectral Angle Mapper (SAM) as one evaluation metric, but this provides only a limited assessment of spectral preservation.

The potential for introducing spectral artifacts or systematic biases through the enhancement process represents a significant concern that requires more thorough investigation. The deep learning approach employed by Sen2-RDSR learns complex nonlinear mappings between input and output imagery, which may inadvertently introduce subtle spectral distortions that could impact downstream applications.

4 Own Implementation Using PyTorch

To better understand the technical details and validate the reproducibility of the Sen2-RDSR approach, a simplified implementation was developed using PyTorch. This section describes the experimental setup, implementation details, and preliminary results obtained from this reproduction effort. [Repository, 2025]

4.1 Implementation Overview

The PyTorch implementation focused on reproducing the core architectural components of Sen2-RDSR (see `utils/Sen2-RDSD.py`). Due to computational constraints and data availability limitations, the implementation was scaled down to work with smaller patch sizes and a reduced number of training samples compared to the original study, the original used 64 big images.

The implementation consisted of several key components: a data preprocessing pipeline for handling Sentinel-2 imagery, the RRDB block implementation, the two-branch network architecture, and training procedures following the two-stage approach described in the original paper.

4.2 Dataset Preparation and Preprocessing

The small dataset was constructed using publicly available Sentinel-2 imagery from the Copernicus Open Access Hub. <https://browser.dataspace.copernicus.eu/> The data split and patches are exactly the as described in the paper.

The preprocessing pipeline to create lower resolution versions of the 10m, 20m and 60m bands. This pipeline was followed using the Dsen2 public repository (<https://github.com/lanha/DSen2/tree/master/training>). [Lanaras et al. 2018]

4.3 Training Procedure and Hyperparameters

The training followed the two-stage approach outlined in the original paper(see training/ training.ipynb). In the first stage, only the 20m band enhancement branch was trained for 100 epochs using L1 loss function. The second stage freezes the weights of the first branch and continue the training with the synthetically generated input from HR, LR and VLR images.

The Adam optimizer was used with an initial learning rate of $1e-4$, which was reduced by $1e-7$ every epoch. Batch size was set to 64 and 16 for 20m and 60m images due to memory constraints, and gradient clipping was applied to stabilize training. The total training time was approximately 3 hours on a single L4 GPU on google colab. Figure .

4.4 Experimental Results and Validation

The preliminary results from the PyTorch implementation showed promising trends consistent with the findings reported in the original paper. Quantitative evaluation using RMSE and SSIM metrics demonstrated improvements over the others results, though the absolute performance values were lower than those reported in the original study due the limited training data and compute.

Figure 7 presents the quantitative results obtained from the PyTorch implementation, comparing the performance against bicubic interpolation and showing the progression of improvements during training. While the absolute values are lower than the original paper due to implementation constraints, the relative improvements demonstrate the effectiveness of the approach.

5 Conclusions

The work by Salgueiro et al. represents a valuable contribution to satellite image super-resolution research, demonstrating both technical innovation and practical improvements over existing methods. The Sen2-RDSR model advances the state-of-the-art until 2021 in Sentinel-2 super-resolution while providing a foundation for future research in this important area. The unified architecture approach, cascaded enhancement strategy, and the evaluation methodology establish new standards for satellite image super-resolution research.

However, the full realization of the potential benefits requires addressing several limitations identified in this analysis, particularly regarding generalization, and spectral fidelity preservation. Future research should focus on developing more robust evaluation frameworks, expanding the geographical diversity of training datasets, and investigating the temporal consistency of enhancement results.

Author contributions. The author conducted the literature review, critical analysis, and real world application examples.

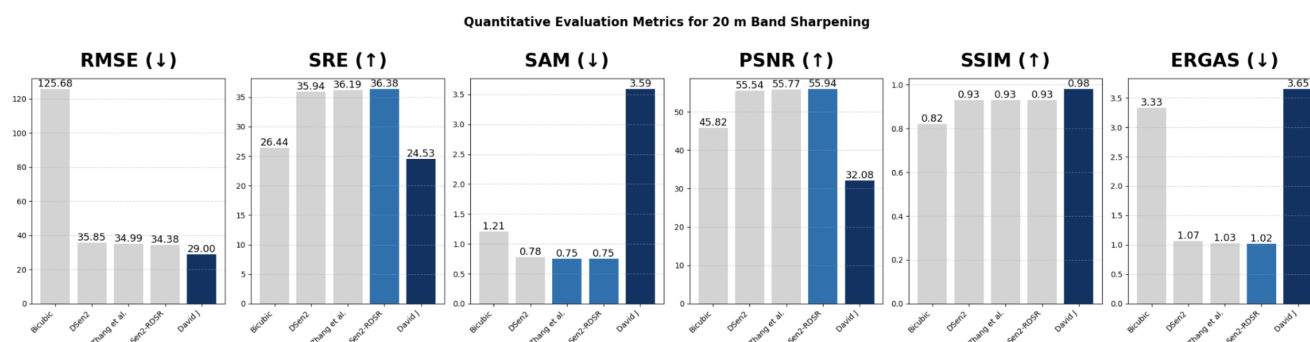


Figure 7. Quantitative results from PyTorch implementation show lower RMSE and higher SSIM than other methods for 20-m band enhancement.

Competing interests. The author declares no competing interests.

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